

संभव BATCH

Geography: Solar System, Rocks, Volcanoes & Geomorphology

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SSC CGL Level Exams | Hard Difficulty | Pure Practice Set (No Answer Key)

Q1. Assertion (A): The Jovian planets have thick atmospheres composed mostly of helium and hydrogen.

Reason (R): The solar wind was less intense in the outer solar system, preventing the primordial atmosphere from being blown off.

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

SSC CGL Tier-II 2022

Q2. Consider the following statements regarding the Solar System:

1. Venus and Uranus rotate on their axes in a retrograde direction (East to West).
2. The orbital velocity of planets increases as their distance from the Sun increases.
3. The Asteroid belt lies between the orbits of Mars and Jupiter.

Which of the statements given above is/are correct?

- (A) 1, 2, and 3
- (B) 2 and 3 only
- (C) 1 and 2 only
- (D) 1 and 3 only

SSC CPO 2021

Q3. Match List-I (Planet) with List-II (Distinct Feature):

List-I

- A. Mercury**
- B. Venus**
- C. Jupiter**
- D. Saturn**

List-II

- 1. Lowest density in the solar system**
- 2. Hottest planet due to runaway greenhouse effect**
- 3. Lacks any atmosphere**
- 4. Possesses the Great Red Spot**

- (A) A-2, B-3, C-4, D-1
- (B) A-3, B-2, C-4, D-1
- (C) A-3, B-4, C-1, D-2
- (D) A-1, B-2, C-4, D-3

SSC CHSL 2022 Pattern

Q4. Which of the following statements about the 'Goldilocks Zone' is NOT correct?

- (A) It refers to the habitable zone around a star.
- (B) It is the region where the temperature is just right for liquid water to exist on a planet's surface.
- (C) Earth is located perfectly within the Sun's Goldilocks Zone.
- (D) It is heavily dependent on the intense solar flares and coronal mass ejections of the star.

SSC Selection Post Phase XI

Q5. Which of the following is the correct chronological sequence of the formation of the universe according to the Big Bang Theory?

- 1. Formation of neutral atoms (Recombination)**
- 2. Inflation and rapid expansion**
- 3. Formation of the first stars and galaxies**
- 4. Nucleosynthesis (formation of first nuclei)**

- (A) 4, 2, 1, 3
- (B) 2, 4, 1, 3
- (C) 2, 1, 4, 3
- (D) 4, 1, 2, 3

SSC CGL 2023

Q6. Assertion (A): Comets develop a glowing tail only when they approach the Sun.
Reason (R): The solar heat vaporizes the comet's ice, and the solar wind blows the gas and dust away from the Sun, forming the tail.

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

SSC MTS 2021

Q7. Match List-I (Moons) with List-II (Planets):

List-I

- A. Ganymede
- B. Titan
- C. Triton
- D. Phobos

List-II

- 1. Mars
- 2. Jupiter
- 3. Saturn
- 4. Neptune

- (A) A-2, B-4, C-3, D-1
- (B) A-2, B-3, C-4, D-1
- (C) A-4, B-3, C-2, D-1
- (D) A-3, B-2, C-4, D-1

SSC CGL Tier-II 2021

Q8. Consider the following statements about the Sun:

- 1. The photosphere is the visible surface of the Sun.
- 2. The corona is much hotter than the photosphere.
- 3. Sunspots are cooler, darker regions on the Sun's surface caused by magnetic activity.

Which of the statements given above are correct?

- (A) 1, 2 and 3
- (B) 2 and 3 only
- (C) 1 and 2 only
- (D) 1 and 3 only

SSC CPO 2022

Q9. Which of the following celestial bodies was demoted from planetary status to a 'dwarf planet' in 2006 by the International Astronomical Union (IAU)?

- (A) Pluto
- (B) Eris
- (C) Ceres
- (D) Makemake

SSC CHSL 2021

Q10. The hypothetical boundary in space beyond which the solar wind is stopped by the interstellar medium is known as:

- (A) Heliopause
- (B) Kuiper Belt
- (C) Roche Limit
- (D) Oort Cloud

SSC CGL Tier-I 2022

Q11. Assertion (A): Igneous rocks are considered the primary or parent rocks.

Reason (R): They are formed directly by the cooling and solidification of magma or lava.

- (A) Both A and R are true but R is NOT the correct explanation of A.
- (B) Both A and R are true and R is the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

SSC Selection Post Phase X

Q12. Match List-I (Rock Type) with List-II (Example):

List-I

- A. Intrusive Igneous**
- B. Extrusive Igneous**
- C. Clastic Sedimentary**
- D. Foliated Metamorphic**

List-II

- 1. Basalt**
- 2. Granite**
- 3. Gneiss**
- 4. Sandstone**

(A) A-2, B-1, C-4, D-3

(B) A-2, B-1, C-3, D-4

(C) A-1, B-2, C-4, D-3

(D) A-4, B-3, C-2, D-1

SSC CGL 2020

Q13. Consider the following statements regarding sedimentary rocks:

- 1. They are formed through the process of lithification.**
- 2. They are the only type of rocks that contain fossils.**
- 3. Limestone and chalk are examples of clastic sedimentary rocks.**

Which of the statements given above is/are NOT correct?

(A) 1 and 2 only

(B) 3 only

(C) 1, 2 and 3

(D) 2 and 3 only

SSC CPO 2023

Q14. In Bowen's Reaction Series, which of the following minerals crystallizes first at the highest temperature from a cooling magma?

(A) Olivine

(B) Quartz

(C) Biotite Mica

(D) Orthoclase Feldspar

SSC CGL Tier-II 2023

Q15. Match List-I (Original Rock) with List-II (Metamorphic Equivalent):

List-I

- A. Limestone**
- B. Sandstone**
- C. Shale**
- D. Granite**

List-II

- 1. Slate**
- 2. Marble**
- 3. Quartzite**
- 4. Gneiss**

(A) A-4, B-3, C-2, D-1

(B) A-2, B-3, C-1, D-4

(C) A-3, B-2, C-1, D-4

(D) A-2, B-1, C-4, D-3

SSC CHSL 2020

Q16. Assertion (A): Basaltic lava is highly fluid and flows over long distances before solidifying.

Reason (R): Basaltic lava is characterized by a low silica content and high temperature.

(A) A is true but R is false.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is false but R is true.

(D) Both A and R are true and R is the correct explanation of A.

SSC MTS 2022

Q17. Which of the following rocks is an example of an organically formed sedimentary rock?

- (A) Sandstone
- (B) Rock Salt
- (C) Coal
- (D) Breccia

SSC CGL 2021

Q18. Consider the following statements about rock textures:

- 1. Extrusive igneous rocks generally have a fine-grained (aphanitic) texture due to rapid cooling.**
- 2. Intrusive igneous rocks have a coarse-grained (phaneritic) texture due to slow cooling deep within the crust.**

Which of the statements given above is/are correct?

- (A) Both 1 and 2
- (B) 1 only
- (C) Neither 1 nor 2
- (D) 2 only

SSC CPO 2021

Q19. Which of the following represents the correct sequence of processes in the Rock Cycle to convert an igneous rock into a metamorphic rock?

- (A) Melting, Cooling, Crystallization
- (B) Weathering, Melting, Lithification
- (C) Weathering, Erosion, Deposition, Lithification, Heat and Pressure
- (D) Heat and Pressure directly without weathering

SSC Selection Post Phase XI

Q20. The term 'Foliation' in geology is exclusively associated with which type of rocks?

- (A) Metamorphic rocks
- (B) Igneous rocks
- (C) Clastic sedimentary rocks
- (D) Chemical sedimentary rocks

SSC CHSL 2022

Q21. Assertion (A): Shield volcanoes are the largest of all volcanoes on Earth but are not extremely explosive.

Reason (R): They are constructed entirely of highly viscous, silicic lava that traps gases.

- (A) Both A and R are true but R is NOT the correct explanation of A.
- (B) A is true but R is false.
- (C) A is false but R is true.
- (D) Both A and R are true and R is the correct explanation of A.

SSC CGL Tier-II 2022

Q22. Match List-I (Volcanic Landforms) with List-II (Description):

List-I

- A. Batholith**
- B. Laccolith**
- C. Dyke**
- D. Sill**

List-II

- 1. Vertical or near-vertical wall-like intrusion**
- 2. Horizontal, sheet-like intrusion**
- 3. Large, deep-seated, dome-shaped magma body**
- 4. Mushroom-shaped intrusion with a pipe-like conduit**

(A) A-4, B-3, C-1, D-2

(B) A-3, B-4, C-2, D-1

(C) A-1, B-2, C-3, D-4

(D) A-3, B-4, C-1, D-2

SSC CPO 2020

Q23. Which of the following statements about Calderas is NOT correct?

- (A) They are the most explosive type of volcanoes.
- (B) Yellowstone and Crater Lake are famous examples of calderas.
- (C) They are formed by the slow, continuous effusion of low-viscosity lava.
- (D) They form when a massive magma chamber empties rapidly, causing the volcano to collapse into itself.

SSC CGL 2023

Q24. Consider the following volcanic features:

- 1. Cinder Cones**
- 2. Composite Volcanoes (Stratovolcanoes)**
- 3. Mid-Ocean Ridge Volcanoes**

Which of the above is/are characterized by alternating layers of lava and pyroclastic material?

- (A) 2 only
- (B) 1 and 2 only
- (C) 1, 2 and 3
- (D) 3 only

SSC MTS 2021

Q25. The Hawaiian Islands are primarily formed by which type of volcanic activity?

- (A) Continental rift volcanism
- (B) Subduction zone volcanism
- (C) Hotspot (Mantle plume) volcanism
- (D) Mid-ocean ridge spreading

SSC CHSL 2019

Q26. Assertion (A): The Pacific 'Ring of Fire' accounts for the majority of the world's active volcanoes and earthquakes.

Reason (R): It represents a continuous series of divergent plate boundaries spanning the Pacific Ocean.

- (A) A is false but R is true.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) Both A and R are true and R is the correct explanation of A.

SSC CGL Tier-II 2020

Q27. Which of the following is the most abundant volcanic gas emitted during an eruption?

- (A) Sulphur dioxide
- (B) Carbon dioxide
- (C) Water vapour
- (D) Hydrogen sulphide

SSC Selection Post Phase X

Q28. Which type of volcanic eruption is characterized by the explosive ejection of highly viscous magma, producing glowing avalanches of hot ash and gas (pyroclastic flows)?

- (A) Hawaiian eruption
- (B) Fissure eruption
- (C) Icelandic eruption
- (D) Pelean eruption

SSC CPO 2022

Q29. Match List-I (Pyroclastic Material) with List-II (Particle Size):

List-I

- A. Volcanic Ash**
- B. Lapilli**
- C. Volcanic Bombs**

List-II

- 1. Larger than 64 mm (partially molten upon ejection)**
- 2. Less than 2 mm**
- 3. Between 2 mm and 64 mm**

- (A) A-2, B-3, C-1
- (B) A-2, B-1, C-3
- (C) A-3, B-2, C-1
- (D) A-1, B-2, C-3

SSC CGL 2021

Q30. A saucer-shaped intrusive igneous body, formed in the syncline or anticline of folded strata, is called a:

- (A) Laccolith
- (B) Batholith
- (C) Phacolith
- (D) Lopolith

SSC CHSL 2021

Q31. Consider the following statements regarding Exogenic and Endogenic processes:

- 1. Endogenic forces are primarily driven by the internal heat of the Earth.**
- 2. Exogenic forces derive their energy from the Earth's atmosphere, which is ultimately driven by solar energy.**
- 3. Weathering and erosion are examples of endogenic processes.**

Which of the statements given above are correct?

- (A) 2 and 3 only
- (B) 1, 2 and 3
- (C) 1 and 3 only
- (D) 1 and 2 only

SSC CGL Tier-I 2022

Q32. Assertion (A): Chemical weathering is more pronounced in hot and humid climates than in cold and dry climates.

Reason (R): High temperature and abundant moisture accelerate chemical reactions such as oxidation, hydration, and carbonation.

- (A) A is false but R is true.
- (B) Both A and R are true and R is the correct explanation of A.
- (C) A is true but R is false.
- (D) Both A and R are true but R is NOT the correct explanation of A.

SSC CPO 2021

Q33. Which of the following is NOT a type of physical (mechanical) weathering?

- (A) Carbonation
- (B) Frost wedging
- (C) Exfoliation
- (D) Thermal expansion and contraction

SSC MTS 2022

Q34. Match List-I (Mass Wasting Type) with List-II (Characteristic):

List-I

- A. Creep**
- B. Solifluction**
- C. Mudflow**
- D. Rockfall**

List-II

- 1. Slow downslope movement of water-saturated soil over permafrost**
- 2. Very slow, imperceptible downhill movement of soil**
- 3. Rapid, free-falling of bedrock from a cliff**
- 4. Rapid flow of debris heavily saturated with water in defined channels**

- (A) A-2, B-1, C-4, D-3
- (B) A-1, B-2, C-3, D-4
- (C) A-2, B-3, C-4, D-1
- (D) A-4, B-1, C-2, D-3

SSC CGL Tier-II 2022 Pattern

Q35. The process by which overlaying rocks are removed by erosion, relieving pressure on the underlying rock and causing it to expand and fracture into sheet-like layers is known as:

- (A) Unloading (Exfoliation)
- (B) Hydration
- (C) Oxidation
- (D) Salt weathering

SSC CHSL 2020

Q36. Consider the following statements about Denudation:

- 1. Denudation is a comprehensive term that includes weathering, mass wasting, and erosion.**
- 2. Weathering is a dynamic process involving the active transportation of materials.**
- 3. Mass wasting is gravity-driven and does not primarily depend on transporting agents like wind or water.**

Which of the statements given above is/are NOT correct?

- (A) 2 only
- (B) 1 and 2 only
- (C) 3 only
- (D) 1 and 3 only

SSC CGL 2023

Q37. In arid regions with limestone formations, the dominant form of weathering is:

- (A) Chemical weathering due to heavy rainfall.
- (B) Biological weathering by deep-rooted desert plants.
- (C) Physical weathering due to temperature changes and wind.
- (D) Carbonation leading to rapid karst topography formation.

SSC Selection Post Phase XI

Q38. Assertion (A): Solifluction is a common mass wasting process in tropical rainforests.

Reason (R): It requires a permanently frozen subsoil (permafrost) which prevents downward percolation of water.

- (A) Both A and R are true but R is NOT the correct explanation of A.
- (B) Both A and R are true and R is the correct explanation of A.
- (C) A is false but R is true.
- (D) A is true but R is false.

SSC CPO 2023

Q39. Which of the following factors does NOT promote mass wasting?

- (A) Seismic activity (earthquakes)
- (B) Over-saturation of soil with water
- (C) Dense vegetation cover on slopes
- (D) Steepening of slopes by natural or human activities

SSC CGL 2019

Q40. Match List-I (Chemical Weathering Process) with List-II (Action):

List-I

- A. Oxidation**
- B. Carbonation**
- C. Hydration**
- D. Solution**

List-II

- 1. Reaction of water with minerals to form a new crystalline structure**
- 2. Reaction of oxygen with minerals, often forming rust in iron-rich rocks**
- 3. Reaction of carbon dioxide and water with minerals like feldspar and limestone**
- 4. Dissolving of soluble minerals directly in water**

- (A) A-1, B-2, C-3, D-4
- (B) A-2, B-3, C-1, D-4
- (C) A-3, B-2, C-1, D-4
- (D) A-2, B-1, C-4, D-3

SSC MTS 2020

Q41. Statement I: Diastrophism is an endogenic geomorphic process that encompasses all processes that move, elevate, or build up portions of the Earth's crust.

Statement II: Volcanism is considered a subset of diastrophic processes.

Which of the following is correct?

- (A) Both statements are incorrect.
- (B) Statement I is correct, Statement II is incorrect.
- (C) Statement I is incorrect, Statement II is correct.
- (D) Both statements are correct.

SSC CHSL 2022

Q42. Which among the following is the defining characteristic of a 'Debris Avalanche'?

- (A) The gradual dissolution of limestone by groundwater.
- (B) Extremely rapid mass movement of unconsolidated rock, soil, and debris.
- (C) Slow, creeping movement of soil over decades.
- (D) A fluid flow of fine-grained mud confined entirely within a river channel.

SSC CGL Tier-II 2020

Q43. Assertion (A): Rivers in their youth stage primarily engage in lateral erosion, widening their valleys.

Reason (R): In the upper course, the gradient is steep, giving the river high kinetic energy predominantly for down-cutting (vertical erosion).

Which of the following is correct?

- (A) Both A and R are true but R is NOT the correct explanation of A.
- (B) A is false but R is true.
- (C) A is true but R is false.
- (D) Both A and R are true and R is the correct explanation of A.

SSC CPO 2021

Q44. Which of the following geomorphic agents is responsible for the formation of 'Mushroom Rocks' (Zeugen)?

- (A) Wind
- (B) Sea waves
- (C) Groundwater
- (D) Glaciers

SSC CGL 2021

Q45. In the context of glacial landforms, what are 'Moraines'?

- (A) Armchair-shaped hollows high on mountainsides formed by glacial erosion.
- (B) Accumulations of dirt and rocks that have fallen onto the glacier surface or have been pushed along by the glacier as it moves.
- (C) Deep, steep-sided inlets of the sea formed by submerged glacial valleys.
- (D) Long, winding ridges of stratified sand and gravel.

SSC Selection Post Phase XI

Q46. Consider the following statements regarding groundwater as a geomorphic agent:

1. Its work is primarily seen in karst topographies.
2. Sinkholes, stalactites, and stalagmites are prominent depositional features.
3. Chemical weathering (solution and precipitation) is its main operational mechanism.

Which of the statements given above is/are NOT correct?

- (A) 1 and 2 only
- (B) 2 only
- (C) 3 only
- (D) 1, 2 and 3

SSC CGL Tier-II 2023

Q47. Match List-I (Geomorphic Agent) with List-II (Landform):

List-I

- A. River (Fluvial)
- B. Glacier
- C. Wind (Aeolian)
- D. Sea Waves

List-II

1. Barchan
2. U-shaped valley
3. Sea Stacks
4. V-shaped valley

- (A) A-3, B-2, C-1, D-4
- (B) A-4, B-2, C-1, D-3
- (C) A-2, B-4, C-1, D-3
- (D) A-4, B-1, C-2, D-3

SSC CHSL 2021

Q48. Which of the following terms describes the breakdown of rocks into smaller particles strictly without any change in their chemical composition?

- (A) Mechanical weathering
- (B) Hydration
- (C) Hydrolysis
- (D) Carbonation

SSC MTS 2022

Q49. Assertion (A): Stalactites grow upwards from the floor of the caves.

Reason (R): They are formed by the precipitation of calcium carbonate from water dripping from the cave ceiling.

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is false but R is true.
- (D) A is true but R is false.

SSC CPO 2022

Q50. What is the primary difference between weathering and erosion?

- (A) Weathering is driven by internal Earth forces, while erosion is driven by external forces.
- (B) Weathering involves the in-situ breakdown of rocks, while erosion involves the transportation of the broken materials.
- (C) Weathering is strictly a chemical process, while erosion is strictly a physical process.
- (D) Weathering occurs only in tropical climates, while erosion occurs globally.

SSC CGL 2020